OpenADMET CYP regression challenge – Method Report

A consensus multitask model was developed using the OCHEM platform.[1] Only data provided by the challenge organisers were used.

The models were generated using DNN[2] and CatBoost[3] ML approaches using default hyperparameters from the OCHEM platform as well as the following descriptors: ALOGPS[4,5] + OESTATE,[6,7] OEstate (60 descriptors that were selected to create solubility model[5]), CDDD,[8] and OSMODRED (https://github.com/osmoai/osmordred).
In addition to descriptor-based methods, Transformer CNF[9,10] & Transformer CNN,[11] and several KGCNN[12] implementations of Graph Neural Networks (AttFP, AttFP trained with MAE as loss function, AttentiveFPPlus, ChemProp and GATv2) representation learning methods were used with default parameters (see Figure 1).

	DNN	SCI:CAT_BOOST	Represent.
ALogPS, OEstate	0.59 0.65 0.47 0.54 (0.5625)	0.58 0.65 0.47 0.56 (0.565)	+
CDDD	0.61 0.66 0.47 0.55 (0.5725)	+	+
OEstate	0.59 0.65 0.47 0.55 (0.565)	+	+
OSMORDRED	0.61 0.66 0.47 0.53 (0.5675)	+	+
CNF2 (F) (3D) 10/10	+	+	0.59 0.64 0.44 0.52 (0.5475)
KGCNN AttFP	+	+	0.61 0.66 0.46 0.53 (0.565)
KGCNN AttFP MAE	+	+	0.6 0.65 0.46 0.52 (0.5575)
KGCNN AttentiveFPPlus	+	+	0.59 0.65 0.45 0.55 (0.56)
KGCNN ChemProp	+	+	0.62 0.65 0.45 0.51 (0.5575)
KGCNN GATv2	+	+	0.6 0.66 0.46 0.53 (0.5625)
TRANSNNI (F) 10/10	+	+	0.58 0.63 0.44 0.51 (0.54)

	Consensus:AVERAGE
Consensus CYP2D6 IC50+CYP1A2 IC50+CYP2C9	0.57 0.62 0.44 0.5 (0.5325)

Figure 1. MAE of individual models developed bagging and their consensus model used for the final prediction.

All these models had low RMSE in Bagging validation, e.g. using a similar protocol (analysis of multiple models and selection best performing in validation) as used in our previous studies.[13,14] All models were averaged (simple average) to create a consensus model, which was used to predict the test set compounds.

The consensus model is available as https://ochem.eu/model/34023962

OpenOCHEM https://github.com/openochem/ contains the code used in this study and the model can be developed and/or exported and used within that GitHub code.

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